

Hand Grenades

PAGES FROM THE ORIGINAL MANUAL

FM 23-30, Hand and Rifle Grenades; Rocket, AT, HE,
2.36-inch (War Department, 1944)

WHAT'S INSIDE

Chapter 1, Section I: every US hand grenade of the time – including the Mk II fragmentation grenade and the M15 white phosphorus grenade – with drawings, fuzes and a data table.

17 pages from the original follow: the manual's cover, then the section about this weapon. Original US Government publication (public domain). Scans courtesy of the Internet Archive and Wikimedia Commons.

FM 23-30

WAR DEPARTMENT BASIC FIELD MANUAL

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HAND AND RIFLE GRENADES

ROCKET, AT, HE,

2.36-INCH

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1942.)

WAR DEPARTMENT • 14 FEBRUARY 1944

(This manual supersedes FM 21-30, 15 June 1942, including C 1, 4 September 1942; C 2, 10 October 1942; C 3, 17 October 1942; C 4, 19 January 1943; C 5, 11 March 1943; C 6, 20 May 1943; C 7, 26 August 1943; Training Circular No. 104, War Department, 1942; Training Circulars Nos. 10, 46, 77, 80, 82, 93, and 120, War Department, 1943.)

Attention is directed to FM 21-7, "List of Training Films, Film Strips, and Film Bulletins," for details as to how appropriate training films and film strips are intended to be used, and how they are made available for training in the use of hand and rifle grenades and the rocket, AT, HE, 2-inch.

CHAPTER 1

HAND GRENADES

SECTION I

GENERAL

1. TYPES. Hand grenades are divided into the following types:

- a.** Fragmentation grenades, containing an explosive charge in a body designed to fragment with the action of the bursting charge.
- b.** Offensive grenades, containing a high explosive charge in a paper body, designed for demolition effect.
- c.** Chemical grenades, containing a chemical agent which produces a toxic or irritant physiological effect, a screening or signal smoke, an incendiary action, or any combination of these.
- d.** Practice grenades, containing a reduced charge, to simulate fragmentation grenades.
- e.** Training grenades, containing no explosive or chemical.

2. FILLERS. The filler is the substance contained in the body of a grenade. Fillers used are —

a. EC (explosive company) blank fire smokeless powder. This is a commercial type granulated nitrocellulose powder, generally pink or yellow in color. It is less powerful than TNT and is exploded by ignition rather than detonation. Grenades loaded with this material can be issued fuze and ready for use and are, in general, not susceptible to mass detonation.

b. Trinitrotoluene (TNT). TNT is stable but, unlike blank fire powder, it explodes by detonation. TNT in block form is used for demolitions.

c. Chloracetophenone (CN). This is a lachrymatory (tear) gas which produces a severe burning sensation in the eyes, causing intense weeping. In one type of irritant hand grenade, CN is combined with diphenylamine-chlorarsine (**d** below).

d. Diphenylaminechlorarsine (DM). DM is a gas which causes a burning sensation in the nose and throat and a heavy or tight feeling in the chest. There is also a nauseating effect, the degree of which depends on the concentration of the gas and the length of exposure.

e. Hexachlorethane-zinc mixture (HC). Upon ignition, HC mixture produces a dense white smoke which is harmless. HC smoke grenades are used by the Army Air Forces and the Armored Command for signals.

f. Sulphur trioxide-chlorosulfonic acid mixture (FS). FS is a corrosive liquid which reacts with the atmosphere, producing an effective screening smoke.

g. Hydrocyanic acid (AC). AC is a clear colorless liquid which vaporizes to a clear colorless gas upon release from the container. It is a powerful nerve poison, nonpersistent, and has an odor of bitter almonds or crushed peaches. A weak concentration causes an unpleasant taste in the mouth, dizziness, headache, and a rush of blood to the head. In strong concentrations, death is instantaneous.

h. Gasoline (thickened) (NP). NP is a thickened incendiary material composed of a mixture of napalm and gasoline. It readily adheres to smooth surfaces and effectively ignites combustible materials.

i. Gasoline (thickened) (IM). IM is a thickened incendiary material which readily adheres to smooth surfaces. It effectively ignites combustible materials.

j. Thermate (TH). TH is an incendiary material composed of thermit and other ingredients which, upon igniting, develops a temperature of 5,000° F. Grenades filled with TH are used to melt together movable metal parts, and against materials which are difficult to ignite.

k. White phosphorus (WP). Upon ignition and contact with the air, white phosphorus burns to a white smoke. Grenades filled with WP are effective agents for screening small unit operations. White phosphorus is also an incendiary; the burning particles produce casualties and have a harassing effect on the enemy.

l. Colored smoke. This is a mixture of sulphur, potassium chlorate, sodium bicarbonate, and dye to produce the desired color. Grenades filled with colored smoke are used as signals for air-ground and ground-to-ground communication.

3. COMPONENTS. a. Time Fuze. (1) FOR DEMOLITIONS AND DESTRUCTION OF DUDS. The time fuze, or safety fuze, is a cord containing a slow-burning powder train. The time fuze issued to the service is known commercially as Bickford fuze and burns at the approximate rate of 15 inches per minute. Time fuze should always be tested before using to determine its rate of burning.

(2) FOR GRENADES. The time fuze in grenade firing mechanisms is cut to burn for varying lengths of time, depending upon the type of grenade. Two fuzes are used with fragmentation grenades: M10A2, cut to burn from

4.3 to 5.5 seconds; M10A3, cut to burn from 4.6 to 4.8 seconds. The M10A2 is commonly referred to as the 4½-second fuze; the M10A3 as the 4-second fuze. Fuzes used with chemical grenades are cut to burn for 2 seconds.

b. Detonators. A detonator is a metal capsule filled with a detonating explosive such as fulminate of mercury. Commercial detonators come in ten sizes, numbered 1 to 10. The higher numbered sizes are larger and contain increasing amounts of the denonating mixture. Nos. 6 and 8 are the ones used in grenades. Detonators are sensitive to heat, shock, and friction and should be handled carefully at all times.

c. Fuzes. The fuze is the mechanism that fires the grenade. Fuzes are described in detail and their functioning explained in TM 9-1985. All hand grenade fuzes are time and automatic. Time means that the grenade is fired after a certain lapse of time and not on percussion. Automatic means that the fuze begins to function automatically as it leaves the hand, providing the safety cotter pin has been removed, thus providing a safety factor by eliminating the necessity of starting the action of the mechanism before the grenade is on its way. As to the final action, fuzes may be classified as detonating or igniting.

(1) **DETONATING FUZES.** Detonating fuzes contain a detonator. The function of the detonator is either to set off the explosive charge or to burst the container and liberate the filler.

(2) **IGNITING FUZES.** Igniting fuzes contain a small quantity of black powder or a powder pellet that ignites the filler as though a lighted match were applied.

(3) **IGNITER M3.** The igniter M3 is used with the frangible grenade M1 (see par. 4a (5)).

4. CHARACTERISTICS. a. The essential characteristics of hand grenades are as follows:

(1) **FRAGMENTATION HAND GRENADE MK. II** (figs. 1 and 2). The body of this grenade is made of cast iron and is about the size of a large lemon. The outside surface is deeply serrated horizontally and vertically to assist in forming uniform fragments when the grenade explodes. The bursting charge is 0.74 ounce of EC blank fire powder, and the grenade is fuzeed with either the M10A2 igniting fuze with a $4\frac{1}{2}$ -second delay or the M10A3 igniting fuze with a 4-second delay. Fragments may fly over 200 yards.

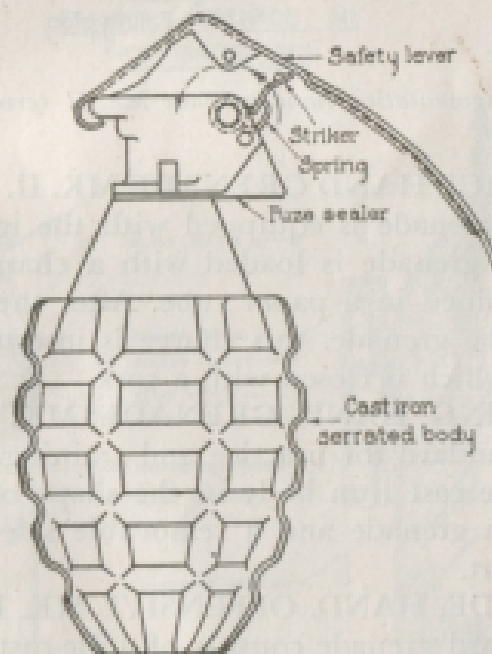


Figure 1. Fragmentation hand grenade Mk. II.

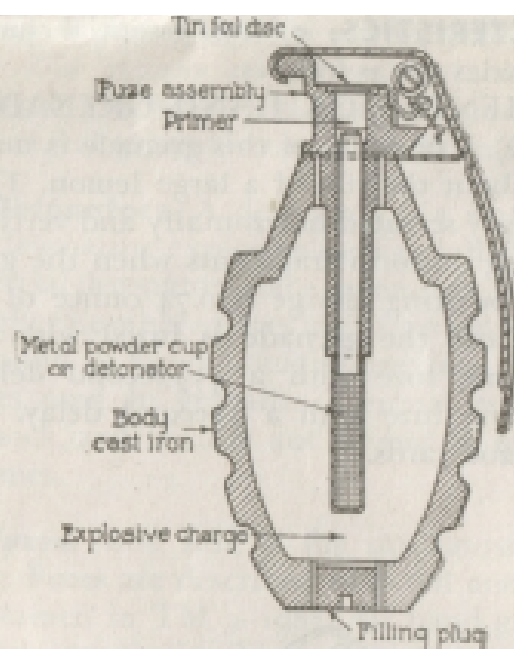


Figure 2. Fragmentation hand grenade Mk. II (cross section).

(2) **PRACTICE HAND GRENADE MK. II.** This standard practice grenade is equipped with the igniting fuze M10A1. The grenade is loaded with a charge of black powder contained in a paper tube. After the fuze is assembled in the grenade, this charge is inserted into the filling hole, which is closed with a cork.

(3) **TRAINING HAND GRENADE MK. 1A1.** This grenade is standard for practice and training. It consists of a one-piece cast iron body in the shape of the fuze fragmentation grenade and a removable safety pin and ring. It is inert.

(4) **GRENADE, HAND, OFFENSIVE MK. IIIA1.** This limited standard grenade consists of a die-cast top which is threaded to receive the fuze, detonating, hand grenade M6A2, and a body of laminated cartridge paper which contains the high-explosive charge. This grenade is for

demolition. It may be used in the open more safely than the fragmentation grenade because there is no marked fragmentation. Grenade bodies and fuzes are shipped separately.

(5) **FRANGIBLE GRENADE M1** (fig. 3). (a) This grenade is a common glass pint bottle equipped with a crimped metal cap and is used with the following chemical fillers: FS, AC, NP, and IM. No bursting charge is provided. Dispersal of contents is produced when the bottle is shattered by impact.

(b) When the frangible grenade M1 is filled with incendiary filler NP or IM, it is equipped with the M3 igniter (par. 3c (3)). This igniter consists of an igniter fuze

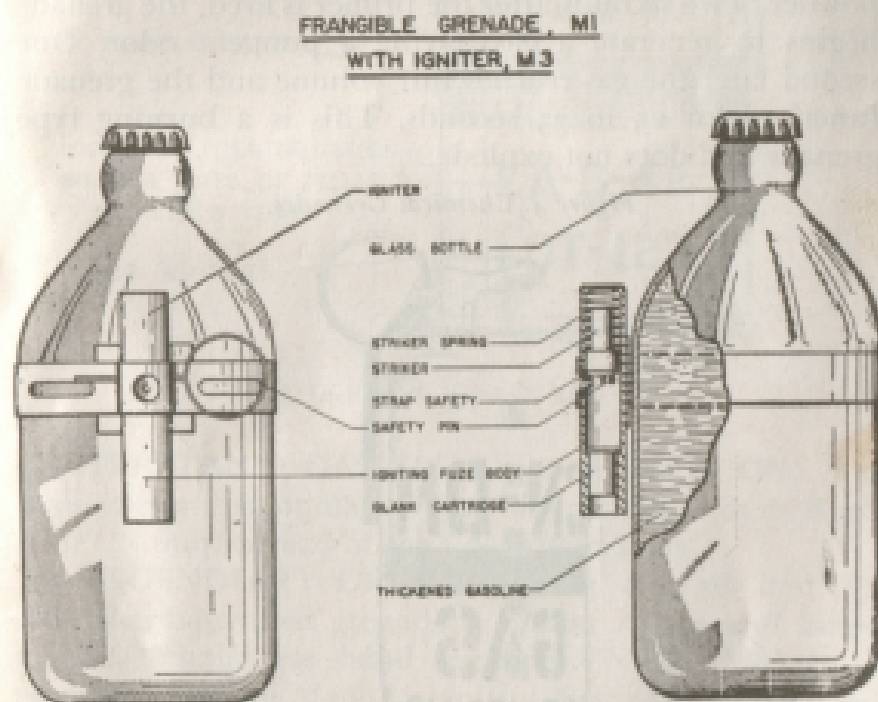
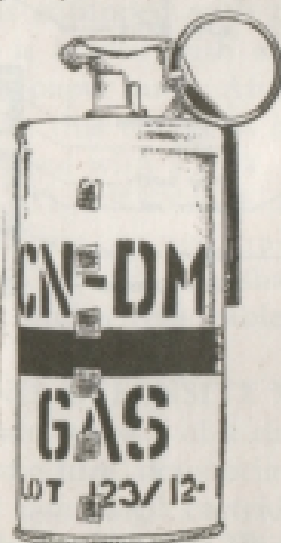


Figure 3. Frangible grenade M1, with igniter M3. (Safety pin ring may be either to the right or the left.)

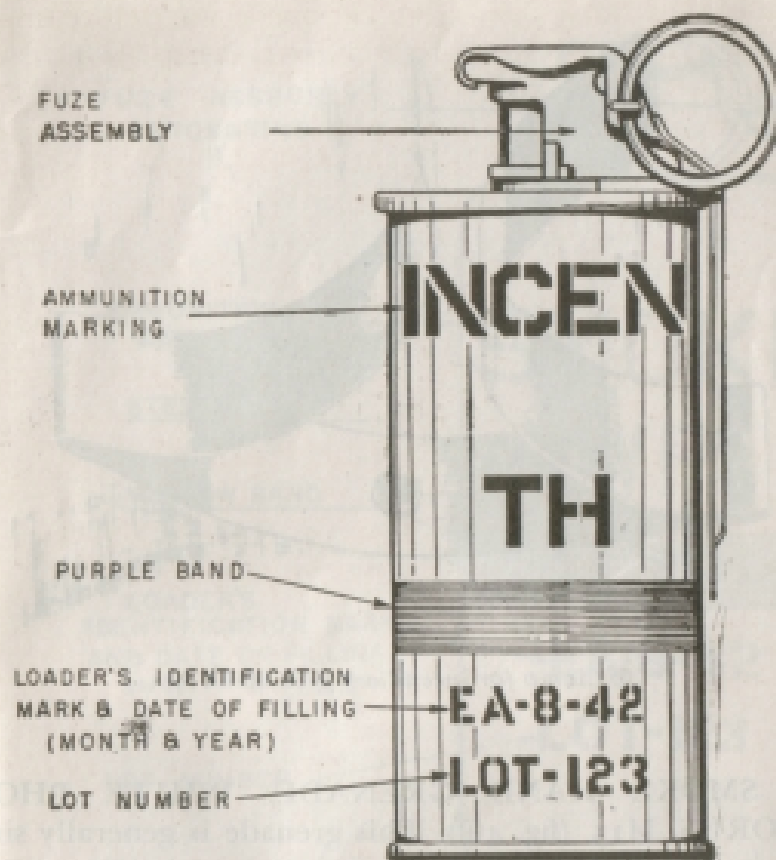
body, a striker, and a blank cartridge, and is fastened around the bottle with a Timmerman strap. Before throwing, the safety pin must be removed. When thrown, the bottle breaks, releasing the strap and strap safety. This permits the striker, which is actuated by the striker spring, to ignite the blank cartridge. The flash from the blank cartridge ignites the contents of the broken bottle.

(6) IRRITANT GAS HAND GRENADE, CN-DM, M6 (fig. 4①). This grenade has a cylindrical body made of tin plate. The body contains perforations or vents which are covered with squares of adhesive tape. When the grenade functions these patches are blown or burned off and the gas escapes. The chemical filler is composed of a mixture of CN-DM and a small amount of blank fire powder. Two seconds after the primer is fired, the grenade begins to generate a gas having a pungent odor. One second later the gas reaches full volume and the grenade functions for 25 to 35 seconds. This is a burning type grenade and does not explode.

Figure 4. Chemical Grenades.



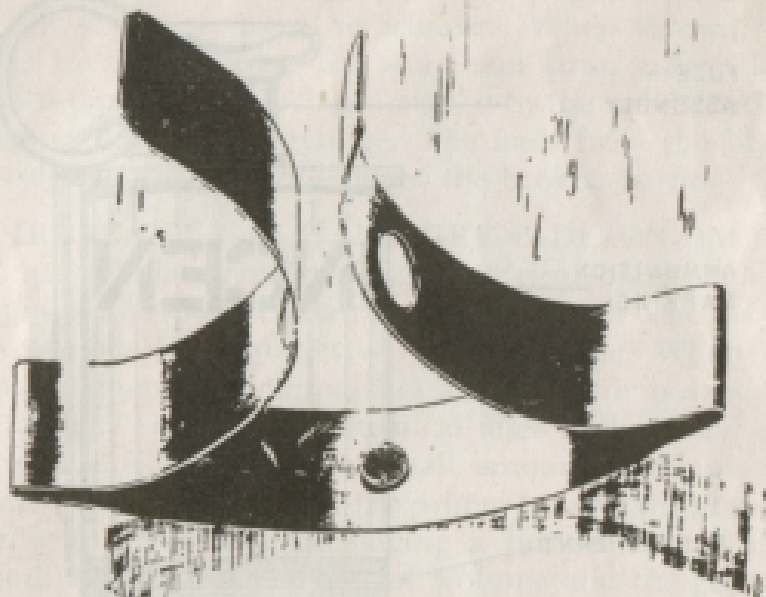
① Irritant gas hand grenade CN-DM M6.



② Incendiary grenade M14.

(7) IRRITANT GAS HAND GRENADE, CN, M7. This grenade is similar to the M6 grenade but is filled with CN only instead of the CN-DM mixture.

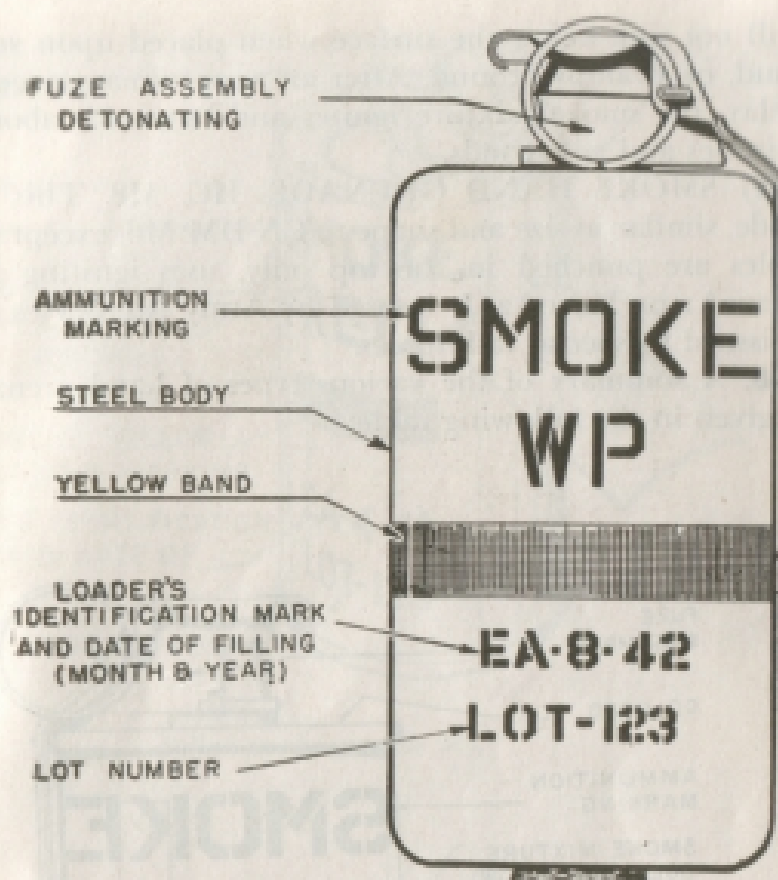
(8) INCENDIARY GRENADE M14 (fig. 4② and ③). This thermate-filled grenade is similar in size and shape to the irritant gas hand grenade CN-DM M6. It is equipped with an M200A1 igniting fuze. After a 2-second delay, the thermate ignites and burns for 30 seconds or longer. A clamp of strap metal is furnished with each grenade, into which the grenade can be placed to keep it from rolling when it ignites.



② Clamp for incendiary grenade M14.

(9) SMOKE HAND GRENADE, WHITE PHOSPHORUS, M15 (fig. 4②). This grenade is generally similar in size and shape to the irritant gas hand grenade CN-DM M6, but has rounded corners. It is equipped with the M6A3 detonating fuze with $4\frac{1}{2}$ -second delay. This grenade is different from the other chemical grenades in that it is a bursting type grenade. On exploding, the solid pieces of WP will be scattered over an area about 25 yards in diameter and will burn for approximately 30 seconds.

(10) COLORED SMOKE HAND GRENADE M18 OR M16 (fig. 4②). This grenade is similar in size and shape to the irritant gas hand grenade CN-DM M6. It is equipped with the M200A1 igniting fuze with 2-second delay. Seven colors of smoke (red, orange, blue, green, black, violet, and yellow) are available. The mixture in



© Smoke hand grenade (WP) M15.

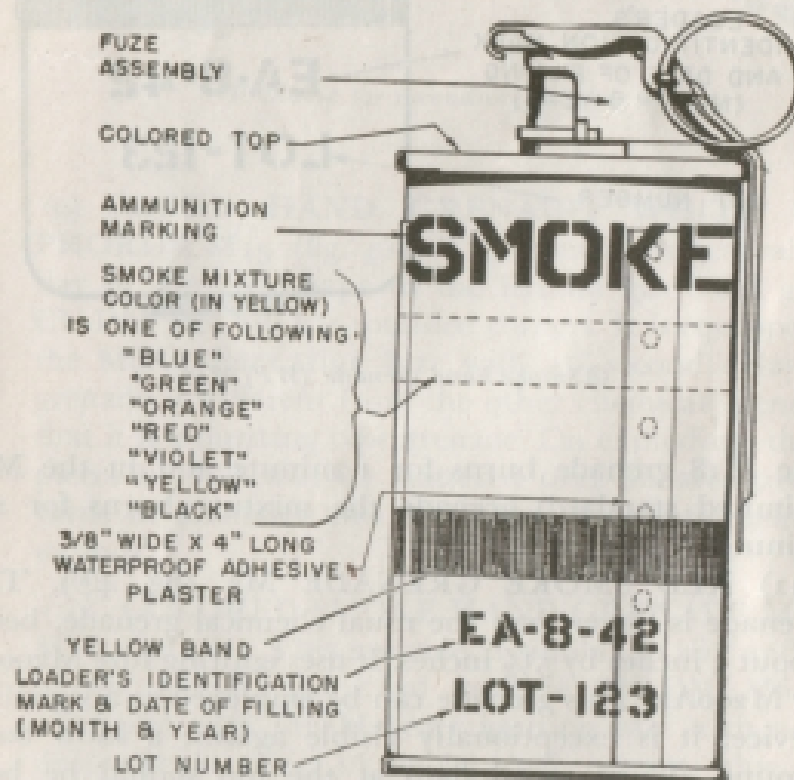
the M18 grenade burns for 1 minute and in the M16 (limited standard) grenade the mixture burns for $2\frac{1}{2}$ minutes.

(11) **RED SMOKE GRENADE M3** (fig. 4©). This grenade is larger than the usual chemical grenade, being about 3 inches by $5\frac{1}{2}$ inches. It uses igniting fuze M200A1 or M200A2. This grenade can be employed as a signaling device; it is exceptionally visible against a snow background. Three metal flaps at the top should be bent upward to serve as additional bases so that the grenade

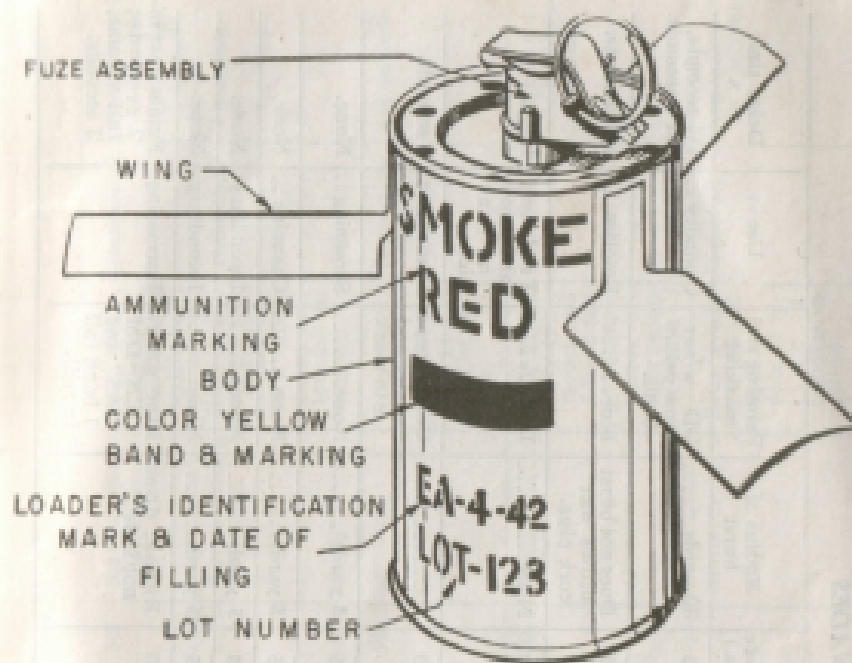
will not sink below the surface when placed upon snow, mud, or swampy ground. After an approximate 2-second delay, the smoke mixture ignites and burns for about 2 minutes and 20 seconds.

(12) SMOKE HAND GRENADE, HC, M8. This grenade, similar in size and shape to CN-DM M6, except that holes are punched in the top only, uses igniting fuze M200A1 or M200A2. It is used by Army Air Forces and is issued to special task forces.

b. A summary of the various types of hand grenades is given in the following table:



© Colored smoke grenade M18 or M16.



© Red smoke grenade M3.

HAND GRENADES

Type	Fuze	Wt. as used (ozs.)	Filler	Body	Range (yds.)	Radius of burst	Painting and marking	Use	Delay time
Grenade, hand, fragmentation Mk. II.	Igniting, M10A2, or M10A3.	20	EC blank fire powder.	Cast iron, serrated.	50	30 yards.	OD w/arrow yellow band.	Casualty ..	4½ seconds, or 4 seconds.
Grenade, hand, practice, Mk. II.	Igniting, M10A1.	20	R e d u c e d charge black powder.	Cast iron, serrated.	50	Does not burst blows out cork plug.	Blue.....	Practice....	4½ seconds.
Grenade, hand, training, Mk. IAI.	None.....	20	None.....	Cast iron, serrated.	50	None.....	Black.....	Training and throwing practice.	Inert.
Grenade, hand, offensive, Mk. IIA1.	Detonating M6A2.	11	TNT.....	Paper....	50		Yellow, type model and lot in black.	Demolition	5 seconds.
Grenade, frangible, M1.	None.....	42	FS.....	Glass.....	25	5 yards.	None.....	Screening..	None.
	None.....	14	AC.....	Glass.....	25	5 yards.	None.....	Casualty ..	None.
	Igniter, M3	24	IM.....	Glass.....	25	5 yards.	None.....	Incendiary.	None.
	Igniter, M3	15	NP.....	Glass.....	25	5 yards.	None.....	Incendiary.	None.
Grenade, hand, gas, irritant, CN-DM, M6.	Igniting M200A1.	17	CN-DM.....	Tinplate, smooth, cylindrical.	50	Burns; does not burst.	Blue gray, "CN-DM GAS" and 1 band (all in red).	Harrassing.	Fuses start in 2 seconds; full volume, 3 seconds.

Grenade, hand, gas, irritant, CN, M7.	Igniting M200A1.	17	CN.....	Tinplate, smooth, cylindri- cal.	50	Burns; does not burst.	Blue gray. "CN GAS" and 1 band (all in red).	Harassing.	Fumes start in 2 seconds; full volume, 3 seconds.
Grenade, incen- dary, M14.	Igniting M200A1.	32	TH.....	Tinplate, smooth, cylindri- cal.		Burns; does not burst.	Blue gray. "TH IN- CENDIA- RY" and 1 band and lot (all in purple).	Incendary.	2 seconds.
Grenade, hand, smoke, WP, M13.	Detonating M6A3.	32	WP.....	Tinplate, cylindri- cal.	50	25 yards.	Blue gray. "WP SMOKE" and 1 band (all in yel- low).	Screening casualty, harassing.	4 1/4 seconds.
Grenade, hand, smoke, col- ored, (red, violet, orange, yellow, black, green, blue). M18 and M16.	Igniting M200A1.	17	Sulphur, po- tassium, chlor- ate, sodium bicarbonate, and dye.	Tinplate, smooth, cylindri- cal.	50	Burns.....	Blue gray. Lot, type, model, and 1 band (all in yellow). Color of filler on top of body.	Signal.....	2 seconds; full volume, 3 seconds.
Grenade, smoke, red, M1.	Igniting M200A1 or M200A2.	23		Steel, cyl- indric.		Burns.....	Blue gray. "SMOKE" RE D, 1 band and 1 band (all in yellow).	Signal.....	Approximate- ly 2 seconds.

HAND GRENADES

Type	Fuze	Wt. as used (ozs.)	Filler	Body	Range (yds.)	Radius of burst	Painting and marking	Use	Delay time
Grenade, hand, smoke, H.C., M8.	Igniting M200A1.	17	H.C. smoke mixture.	Tinplate, smooth, cylindri- cal.	50	Bursts; does not burst.	Blue gray. "HC SMOKE" and 1 band and lot (all in yellow)	Signal-----	Fumes start in 2 seconds; full volume, 3 seconds.

1 Issued fuze ready for use.

2 Vapor will travel a considerable distance down wind.